



Chapter 10

Kinetics: Rates and Mechanisms of Chemical Reactions



Kinetics

- Studies the rate at which a chemical process occurs.
- Besides information about the speed at which reactions occur, kinetics also sheds light on the reaction mechanism (exactly how the reaction occurs).



Outline

- **Reaction Rates**

How we measure rates.

- **Rate Laws**

How the rate depends on amounts of reactants.

- **Integrated Rate Laws**

How long it takes to react 50% of reactants.

- **Arrhenius Equation**

How rate constant changes with T.

- **Mechanisms**

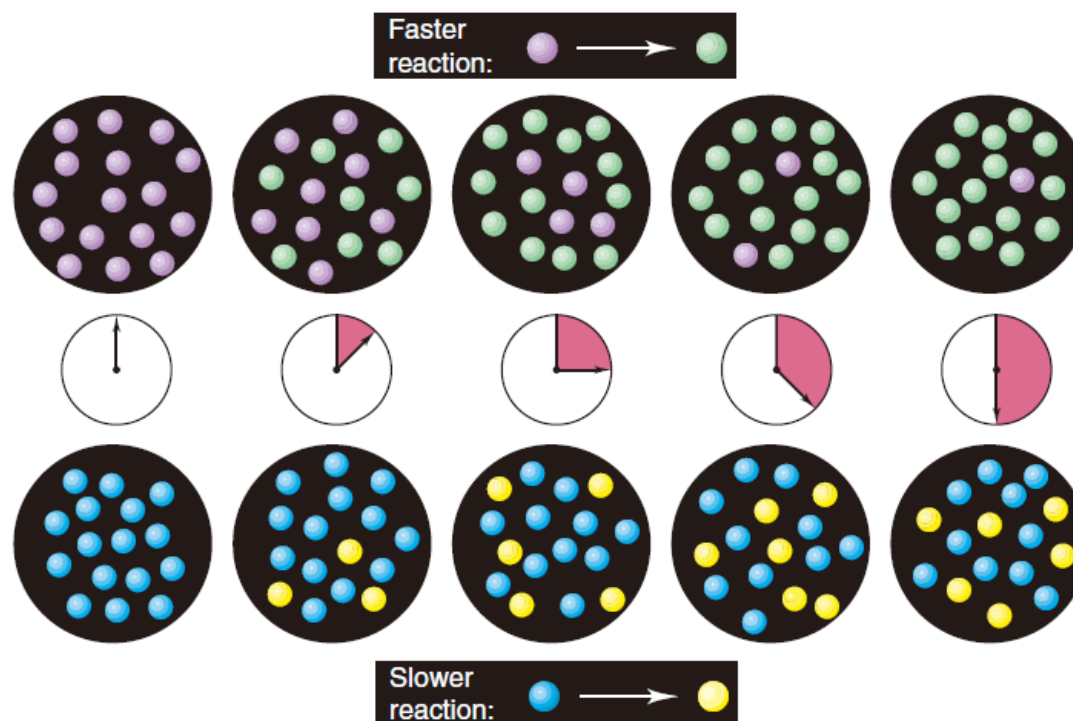
Link between rate and molecular scale processes.



10.1 Reaction rate

Reaction rate

- The change in the concentrations of reactants (or products) as a function of time.
- in a faster reaction (higher rate), the reactant concentration decreases quickly, whereas in a slower reaction (lower rate), it decreases slowly.





Factors That Affect Reaction Rates

- Concentration

Molecules must collide to react

$$\text{Rate} \propto \text{collision frequency} \propto \text{concentration}$$

- Physical state

Molecules must mix to collide

The more finely divided a solid or liquid reactant, the greater its surface area, the more contact it makes with the other reactant, and the faster the reaction occurs.



- Temperature

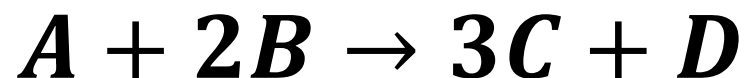
Molecules must collide with enough energy

$$\text{Rate} \propto \text{collision frequency} \propto \text{temperature}$$

$$\text{Rate} \propto \text{collision energy} \propto \text{temperature}$$



10.2 Expressing the reaction rate



The volume/surface area of the reaction system is constant.

At some instant:

$$v = \frac{d[D]}{dt} = \frac{1}{3} \frac{d[C]}{dt} = -\frac{d[A]}{dt} = -\frac{1}{2} \frac{d[B]}{dt} \quad v: \text{the rate of reaction}$$

For a homogeneous reaction

$$v = \frac{1}{v_j} \times \frac{1}{V} \frac{dn_j}{dt} = \frac{1}{v_j} \frac{d[J]}{dt}$$

V : the volume of the system

$$[J] = n_j / V$$

$\text{mol dm}^{-3} \text{s}^{-1}$

For a heterogeneous reaction

$$v = \frac{1}{v_j} \times \frac{1}{A} \frac{dn_j}{dt} = \frac{1}{v_j} \frac{d\sigma_j}{dt}$$

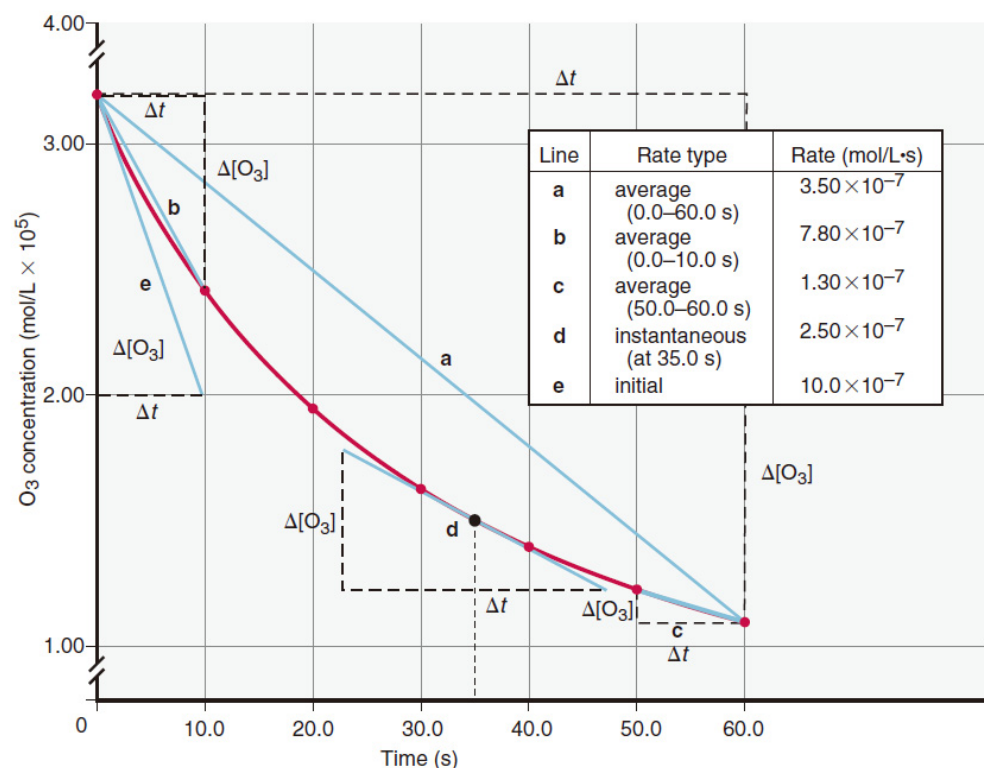
S : the surface area

$$\sigma_j = n_j / A$$

$\text{mol m}^{-2} \text{s}^{-1}$



Average, Instantaneous, and Initial reaction rates

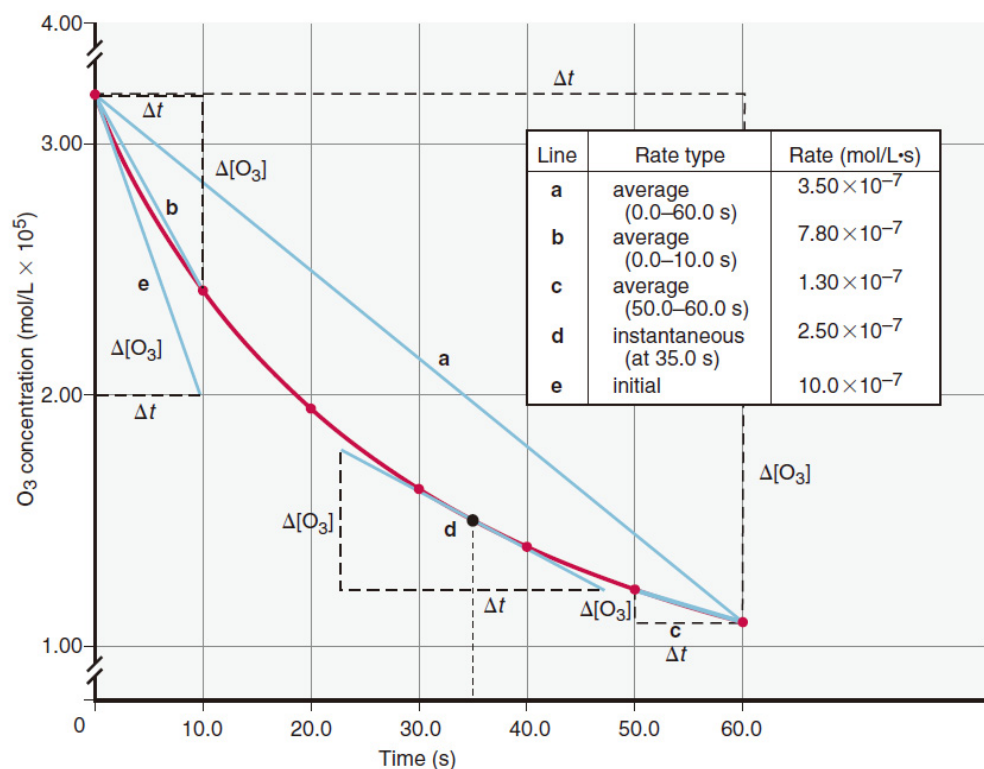


Average rate

Over a given period of time, the average rate is the slope of the line joining two points along the curve.

$$0-10 \text{ s} \quad \text{Rate} = -\frac{\Delta[\text{O}_3]}{\Delta t} = -\frac{(2.42 \times 10^{-5} \text{ mol/L}) - (3.20 \times 10^{-5} \text{ mol/L})}{10.0 \text{ s} - 0.0 \text{ s}} = 7.80 \times 10^{-7} \text{ mol/L} \cdot \text{s}$$

$$50-60 \text{ s} \quad \text{Rate} = -\frac{\Delta[\text{O}_3]}{\Delta t} = -\frac{(1.10 \times 10^{-5} \text{ mol/L}) - (1.23 \times 10^{-5} \text{ mol/L})}{60.0 \text{ s} - 50.0 \text{ s}} = 1.30 \times 10^{-7} \text{ mol/L} \cdot \text{s}$$



Instantaneous rate

The rate at a particular instant during the reaction.

The rate at 35.0 s is:
 $2.50 \times 10^{-7} \text{ mol/L} \cdot \text{s}$

Initial rate

The instantaneous rate at the moment the reactants are mixed (that is, at $t = 0$).

For the initial rate, $t = 0$, so product concentrations are negligible, and so is the reverse rate.



Rate laws and rate constants

A rate law shows the relationship between the reaction rate and the concentrations of reactants.

For a general reaction occurring at a fixed temperature:



The rate law is:

$$v = k[A]^m [B]^n \dots$$

k : rate constant

- Specific for a given reaction at a given temperature: Dependent on T
- Does not change as the reaction proceeds.
- Be found by experiment.

m, n : reaction orders



A reaction has an individual order “with respect to” or “in” each reactant, and an overall order, the sum of the individual orders.

$$v = k[A] [B]$$

First-order in A and first-order in B;
The overall order is second-order.

$$v = k[A]^m [B]^n \dots$$

m-order in A and n-order in B;
The overall order is (m+n)-order.

$$v = k[A]^{1/2} [B]$$

Half-order in A and first-order in B;
The overall order is three-halves-order.

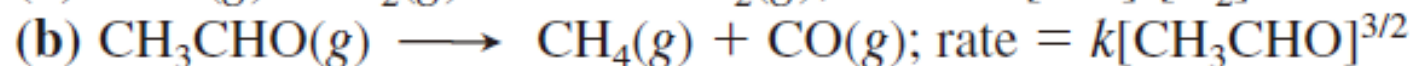
$$v = k$$

A zero-order reaction.



Self test 10.1

Problem For each of the following reactions, use the given rate law to determine the reaction order with respect to each reactant and the overall order:



Answer:

(a) The exponent of $[\text{NO}]$ is 2, so the reaction is second order with respect to NO, first order with respect to O_2 , and third order overall.

(b) The reaction is 3/2 order in CH_3CHO and 3/2 order overall.

(c) The reaction is first order in H_2O_2 , first order in I^- , and second order overall. The reactant H^+ does not appear in the rate law, so the reaction is zero order in H^+ .



10.3 Integrated rate laws

The rate laws answer:

“How fast is the reaction proceeding at the moment y moles per liter of A are mixed with z moles per liter of B?”

The integrated rate laws will answer:

“How long will it take to use up x moles per liter of A?”

“What is $[A]$ after y minutes of reaction?”



(a) First-order reactions

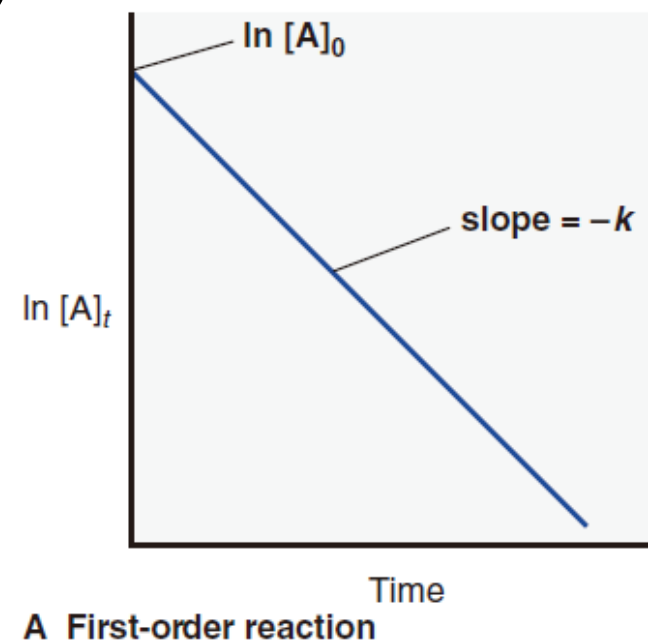


$$\frac{d[A]}{dt} = -k[A] \quad \Longrightarrow \quad \frac{d[A]}{[A]} = -k dt$$

$$\int_{[A]_0}^{[A]} \frac{d[A]}{[A]} = -k \int_0^t dt \quad [A]_0 \text{ is the initial concentration of A (at } t = 0)$$

$$\ln \left(\frac{[A]}{[A]_0} \right) = -kt \quad [A] = [A]_0 e^{-kt}$$

$$\ln[A] = \ln[A]_0 - kt$$





Half-life

- Half-life is defined as the time required for one-half of a reactant to react.
- Because $[A]$ at $t_{1/2}$ is one-half of the original $[A]$,
 $[A]_t = 0.5 [A]_0$.

For the first-order reaction:

$$kt_{1/2} = -\ln\left(\frac{\frac{1}{2}[A]_0}{[A]_0}\right) = -\ln\frac{1}{2} = \ln 2$$

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

For a first-order reaction, the half-life of a reactant is independent of its initial concentration.



(b) Second-order reactions

Type I



$$\frac{d[A]}{dt} = -k[A]^2 \quad \Longrightarrow \quad \frac{d[A]}{[A]^2} = -k dt$$

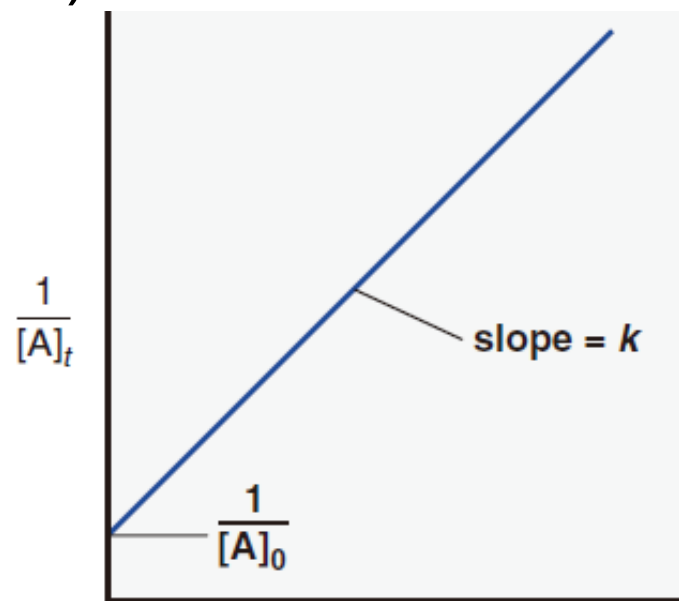
$$\int_{[A]_0}^{[A]} \frac{d[A]}{[A]^2} = -k \int_0^t dt$$

$[A]_0$ is the initial concentration of A (at $t = 0$)

$$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$$

$$[A] = \frac{[A]_0}{1 + kt[A]_0}$$

$$t_{1/2} = \frac{1}{k[A]_0}$$



B Second-order reaction



(b) Second-order reactions

Type II



$$\frac{d[A]}{dt} = -k[A][B]$$

If the reaction is $A + B \rightarrow P$

It follows from the reaction stoichiometry that, when the concentration of A has fallen to $[A]_0 - x$, the concentration of B will have fallen to $[B]_0 - x$ (because each A that disappears entails the disappearance of one B).

$$\frac{d[A]}{dt} = -k([A]_0 - x)([B]_0 - x)$$

Because $[A] = [A]_0 - x$, it follows that $d[A]/dt = -dx/dt$ and the rate law may be written as

$$\frac{dx}{dt} = k([A]_0 - x)([B]_0 - x)$$



$$\int_0^x \frac{dx}{([A]_0 - x)([B]_0 - x)} = k \int_0^t dt$$

To use the method of partial fractions to evaluate an integral of the form

$\int \frac{1}{(a-x)(b-x)} dx$, where a and b are constants, we write

$$\frac{1}{(a-x)(b-x)} = \frac{1}{b-a} \left(\frac{1}{a-x} - \frac{1}{b-x} \right)$$

and integrate the expression on the right. It follows that

$$\begin{aligned} \int \frac{dx}{(a-x)(b-x)} &= \frac{1}{b-a} \left[\int \frac{dx}{a-x} - \int \frac{dx}{b-x} \right] \\ &= \frac{1}{b-a} \left(\ln \frac{1}{a-x} - \ln \frac{1}{b-x} \right) + \text{constant} \end{aligned}$$

$$\int_0^x \frac{dx}{([A]_0 - x)([B]_0 - x)} = \frac{1}{[B]_0 - [A]_0} \left\{ \ln \left(\frac{[A]_0}{[A]_0 - x} \right) - \ln \left(\frac{[B]_0}{[B]_0 - x} \right) \right\}$$

$$\ln \left(\frac{[B]/[B]_0}{[A]/[A]_0} \right) = ([B]_0 - [A]_0)kt$$



(c) Zero-order reactions

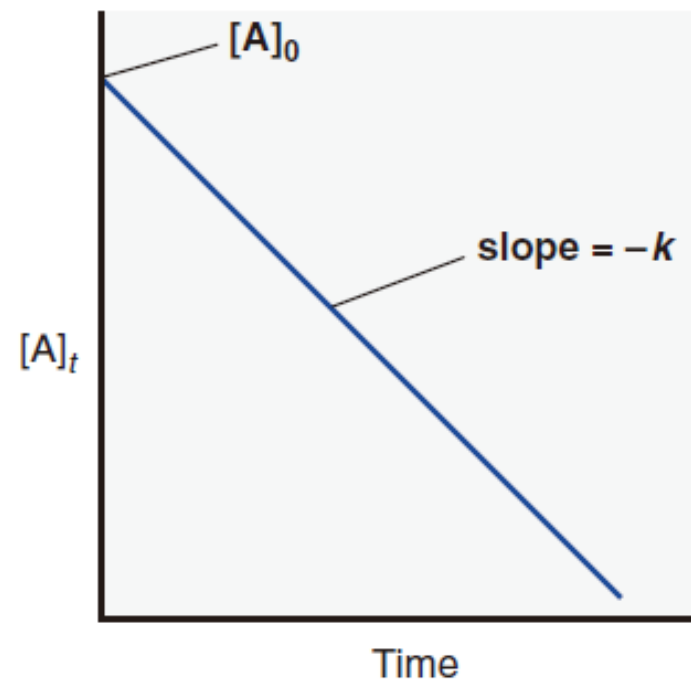


$$-\frac{d[A]}{dt} = k$$

$$\int_{[A]_0}^{[A]} d[A] = -k \int_0^t dt$$

$$[A] = [A]_0 - kt$$

$$t_{1/2} = \frac{[A]_0}{2k}$$



C Zero-order reaction



Order	Reaction	Rate law	Integrated rate law	$t_{1/2}$
0	$A \rightarrow P$	$v = k$	$[A] = [A]_0 - kt$	$\frac{[A]_0}{2k}$
1	$A \rightarrow P$	$v = k[A]$	$\ln\left(\frac{[A]}{[A]_0}\right) = -kt$ $[A] = [A]_0 e^{-kt}$	$\frac{\ln 2}{k} = \frac{0.693}{k}$
2	$A \rightarrow P$	$v = k[A]^2$	$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$ $[A] = \frac{[A]_0}{1 + kt[A]_0}$	$\frac{1}{k[A]_0}$
	$A + B \rightarrow P$	$v = k[A][B]$	$\ln\left(\frac{[B]/[B]_0}{[A]/[A]_0}\right) = ([B]_0 - [A]_0)kt$	



Self test 10.3

The rate constant for the first-order decomposition of a compound A in the reaction $2 A \rightarrow P$ is $k = 2.78 \times 10^{-7} \text{ s}^{-1}$ at 25°C . What is the half-life of A? What will be the pressure, initially 32.1 kPa, at (a) 10 h, (b) 50 h after initiation of the reaction?

$$-\frac{1}{2} \frac{d[A]}{dt} = k[A]$$

$$\frac{d[A]}{dt} = -2k[A]$$

$$\int_{[A]_0}^{[A]} \frac{d[A]}{[A]} = -2k \int_0^t dt$$

$$\ln \left(\frac{[A]}{[A]_0} \right) = -2kt$$

$$[A] = [A]_0 e^{-2kt}$$



$$kt_{1/2} = -\frac{1}{2} \ln \left(\frac{\frac{1}{2} [A]_0}{[A]_0} \right) = -\frac{1}{2} \ln \frac{1}{2} = \frac{1}{2} \ln 2$$

$$t_{1/2} = \frac{1}{2} \frac{\ln 2}{k} = \frac{1}{2} \times \frac{0.693}{2.78 \times 10^{-7} \text{ s}^{-1}} = 1.25 \times 10^6 \text{ s} \approx 347.2 \text{ h}$$

$$[A] = [A]_0 e^{-2kt}$$

$$[A]_{3.6 \times 10^4 \text{ s}} = 32.1 \text{ kPa} \times e^{(-2 \times 2.78 \times 10^{-7} \text{ s}^{-1} \times 3.6 \times 10^4 \text{ s})} = 31.5 \text{ kPa}$$

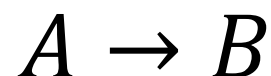
$$[A]_{1.8 \times 10^5 \text{ s}} = 32.1 \text{ kPa} \times e^{(-2 \times 2.78 \times 10^{-7} \text{ s}^{-1} \times 1.8 \times 10^5 \text{ s})} = 29.0 \text{ kPa}$$



10.4 Reactions approaching equilibrium

In practice, most kinetic studies are made on reactions that are far from equilibrium, and the reverse reactions are unimportant.

When the reaction is close to equilibrium, the products may be so abundant that the reverse reaction must be taken into account.



$$v_{forward} = k[A]$$

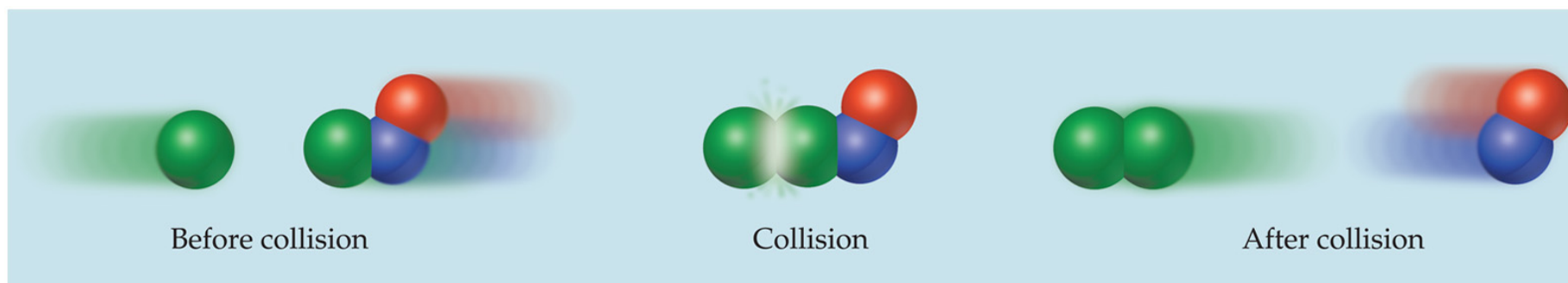
$$v_{reverse} = k'[B]$$



10.5 Theories of chemical kinetics

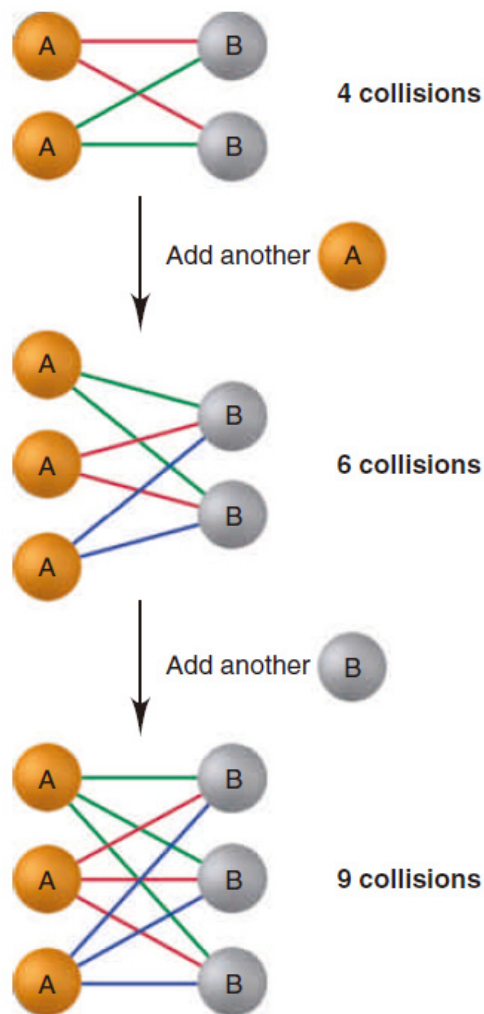
Collision theory: basis of the rate law

- In a chemical reaction, bonds are broken and new bonds are formed.
- Molecules can only react if they collide with each other.
- Furthermore, molecules must collide with the correct orientation and with enough energy to cause bond breakage and formation.





Why concentrations are multiplied in the rate law?



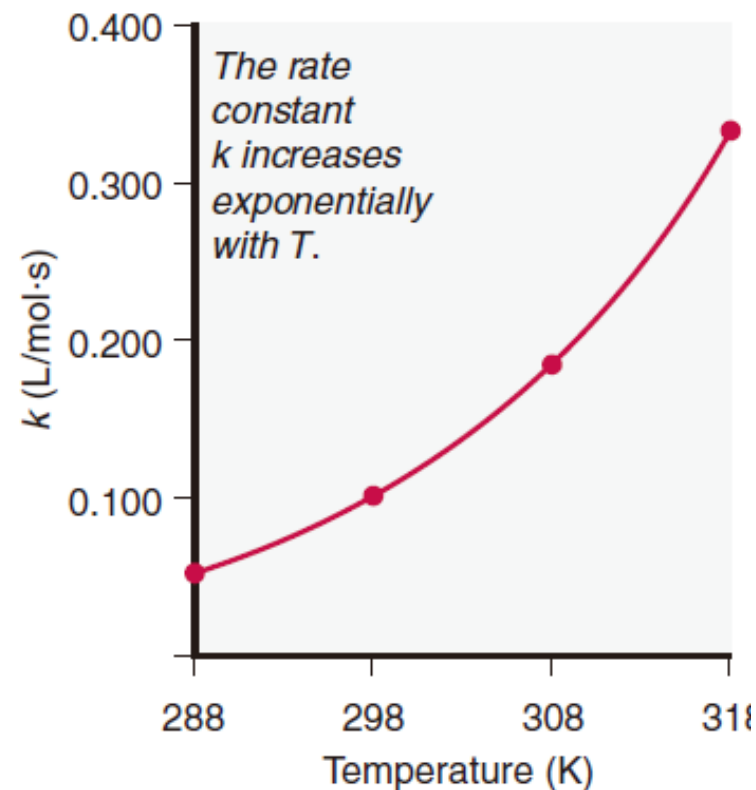
Particles must collide to react, so the collision frequency, the number of collisions per unit time, provides an *upper limit* on how fast a reaction can take place.

Figure 16.13 The number of possible collisions is the product, not the sum, of reactant concentrations.



The Effect of Temperature on the Rate Constant and the Rate

Expt	[Ester]	[H ₂ O]	T (K)	Rate (mol/L·s)	k (L/mol·s)
1	0.100	0.200	288	1.04×10^{-3}	0.0521
2	0.100	0.200	298	2.02×10^{-3}	0.101
3	0.100	0.200	308	3.68×10^{-3}	0.184
4	0.100	0.200	318	6.64×10^{-3}	0.332



Arrhenius equation

$$k = Ae^{-E_a/RT}$$

E_a : activation energy

A : frequency factor

As T increases, the value of the negative exponent becomes smaller, which means k becomes larger, so the rate increases:

Higher $T \Rightarrow$ larger $k \Rightarrow$ increased rate



Activation Energy (E_a)

- Activation energy (E_a), a minimum amount of energy required for reaction.
- Just as a ball cannot get over a hill if it does not roll up the hill with enough energy, a reaction cannot occur unless the molecules possess sufficient energy to get over the activation energy barrier.





Temperature Effect

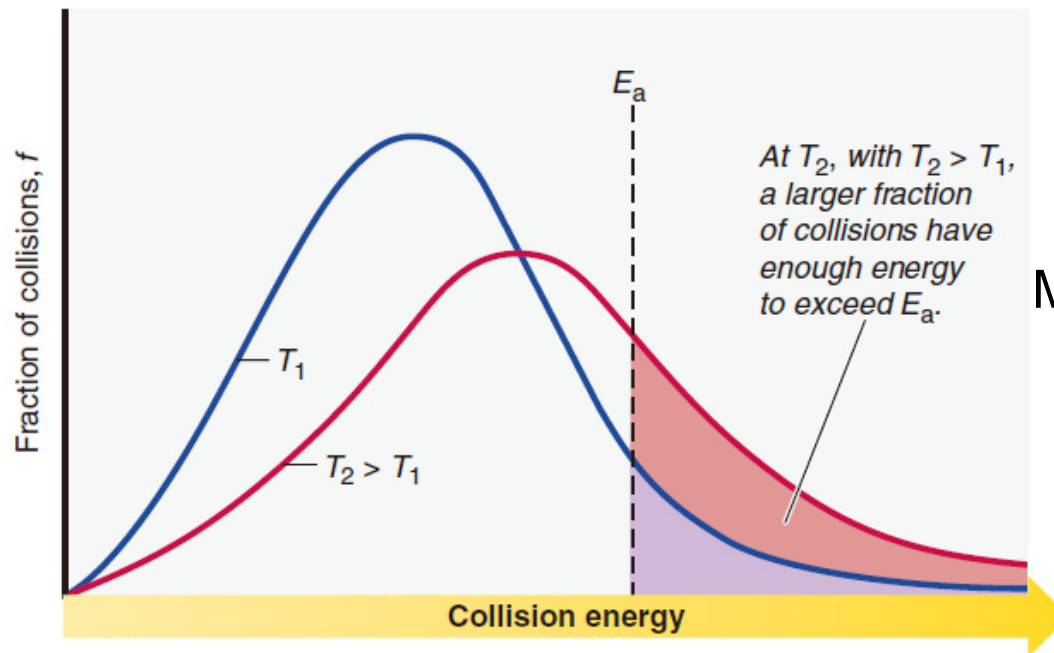
A rise in temperature has two effects on moving particles:

- Causing a higher collision frequency (**minor effect**)

If particles move faster, they collide more often.

- A higher collision energy (**major effect**)

A rise in temperature enlarges the fraction of collisions with enough energy to exceed E_a .



Maxwell-Boltzmann Distributions



The effect of molecular structure on rate

Effective collisions

Those that actually lead to product because the atoms that become bonded in the product make contact. Thus, to be effective, a collision must have **enough energy** and the **appropriate molecular orientation**.

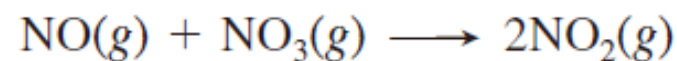
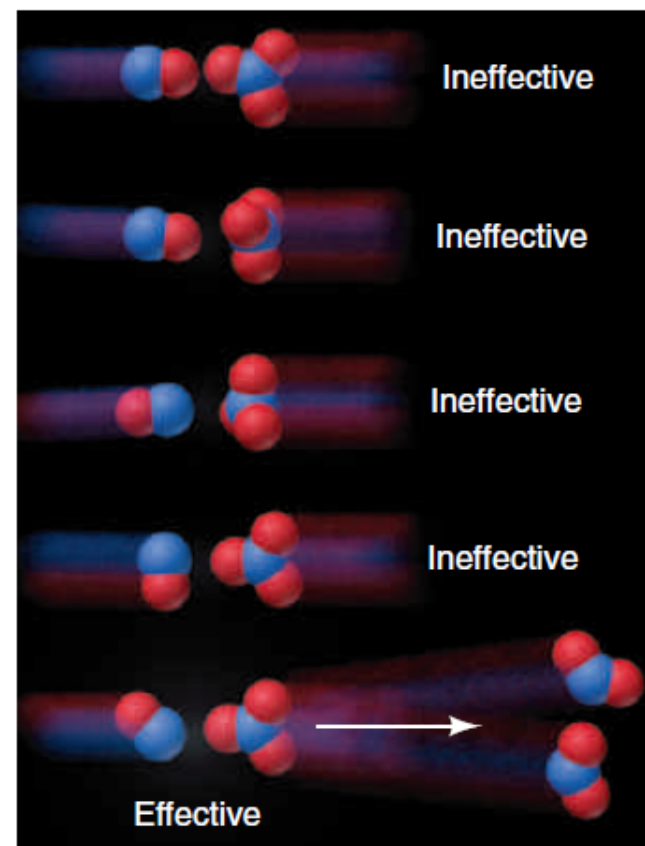
$$k = Ae^{-E_a/RT} \quad A: \text{frequency factor}$$

$$A = pZ$$

Z: collision frequency

p: an orientation probability factor

The more complex the molecular structure, the smaller the p value is.



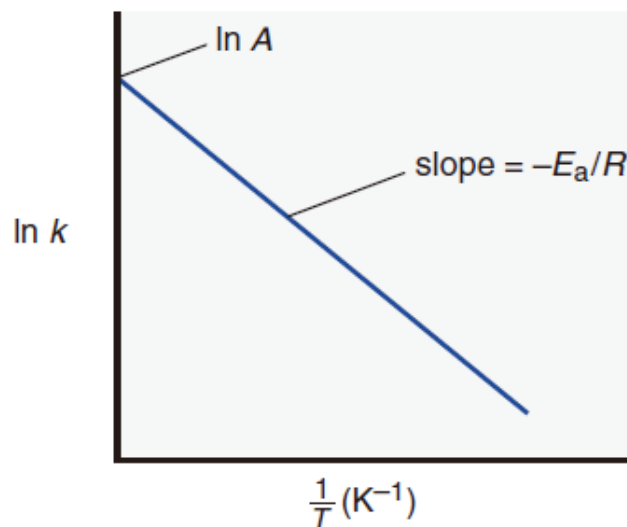
The *p* value for this reaction is 0.006



Calculating the activation energy

$$k = Ae^{-E_a/RT}$$

Method I



$$\ln k = \ln A - \frac{E_a}{R} \left(\frac{1}{T} \right)$$

When k is determined experimentally at several temperatures, E_a can be calculated from the slope of a plot of $\ln k$ vs. $1/T$.

Method II

$$\ln k_1 = \ln A - \frac{E_a}{R} \left(\frac{1}{T_1} \right)$$

$$\ln k_2 = \ln A - \frac{E_a}{R} \left(\frac{1}{T_2} \right)$$

$$\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

We can find E_a in another way if we know the rate constants at two temperatures.



Self test 10.4

The rate constant for the formation of hydrogen iodide from its elements is $2.7 \times 10^{-4} \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$ at 600 K and $3.5 \times 10^{-3} \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$ at 650 K. Find the activation energy, E_a .



$$\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\ln \frac{3.5 \times 10^{-3}}{2.7 \times 10^{-4}} = -\frac{E_a}{8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}} \left(\frac{1}{650} - \frac{1}{600} \right)$$

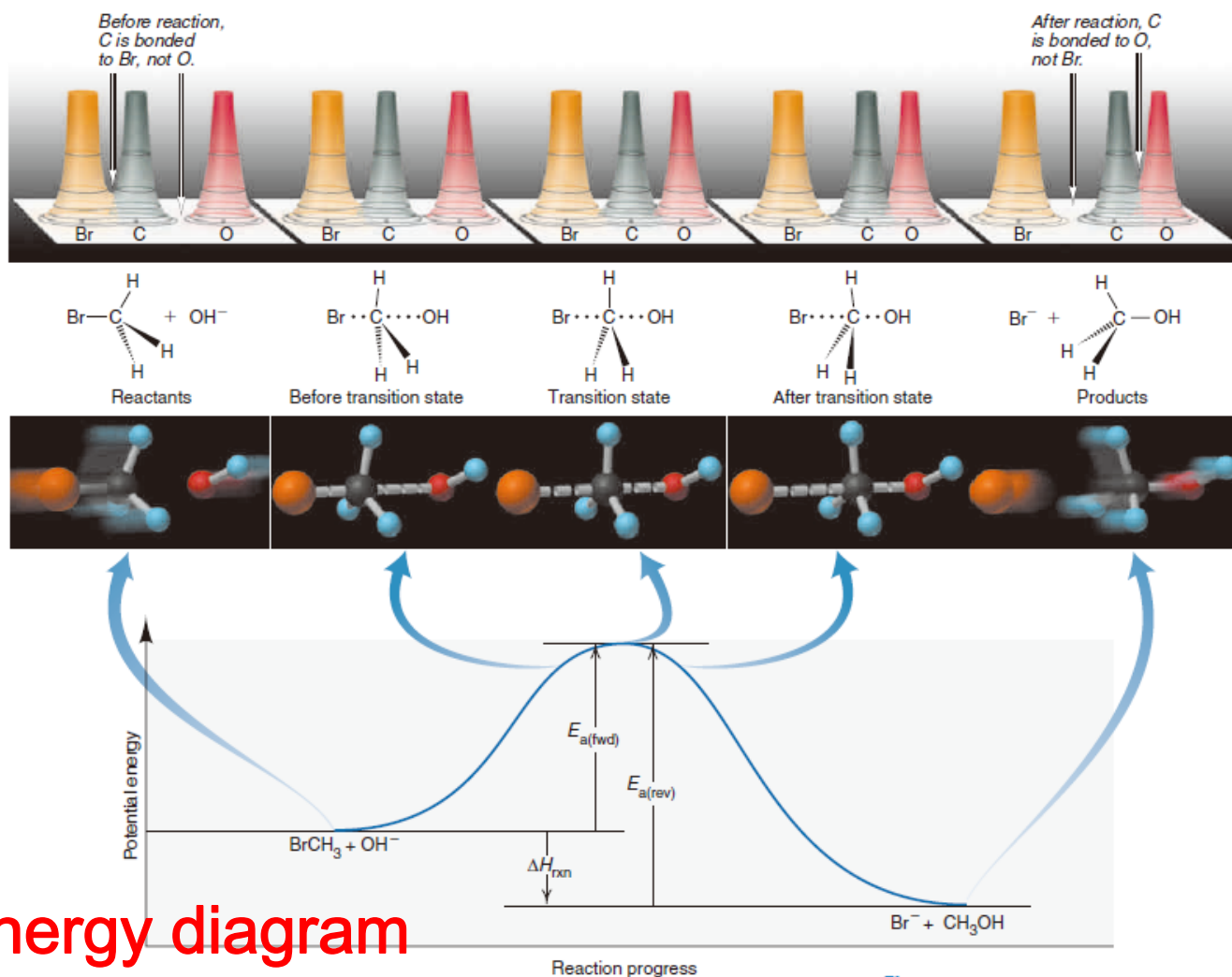
$$E_a = 1.66 \times 10^5 \text{ J}$$



Transition state theory

At some point during this smooth transformation, a species with partial bonds exists that is neither reactant nor product.

Transition state exists only at the instant of highest potential energy. Thus, the activation energy of a reaction is used to reach the transition state.



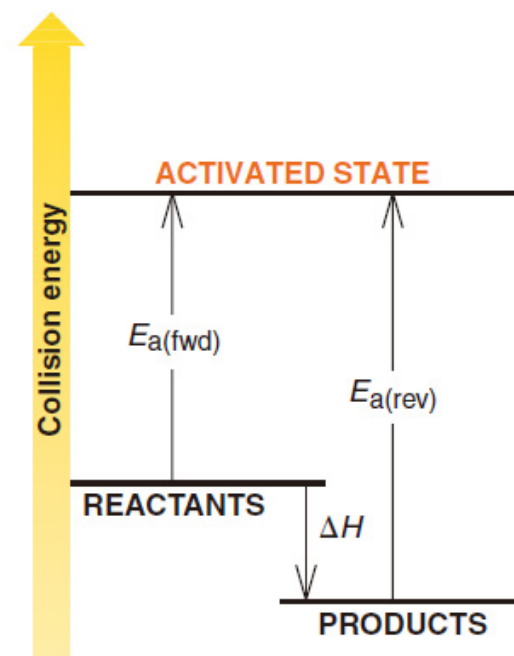


$E_{a(fwd)}$ and $E_{a(rev)}$

- The activation energy for the forward reaction, $E_{a(fwd)}$, is the energy difference between the transition state and the reactants;
- The activation energy for the reverse reaction, $E_{a(rev)}$, is the energy difference between the transition state and the products.

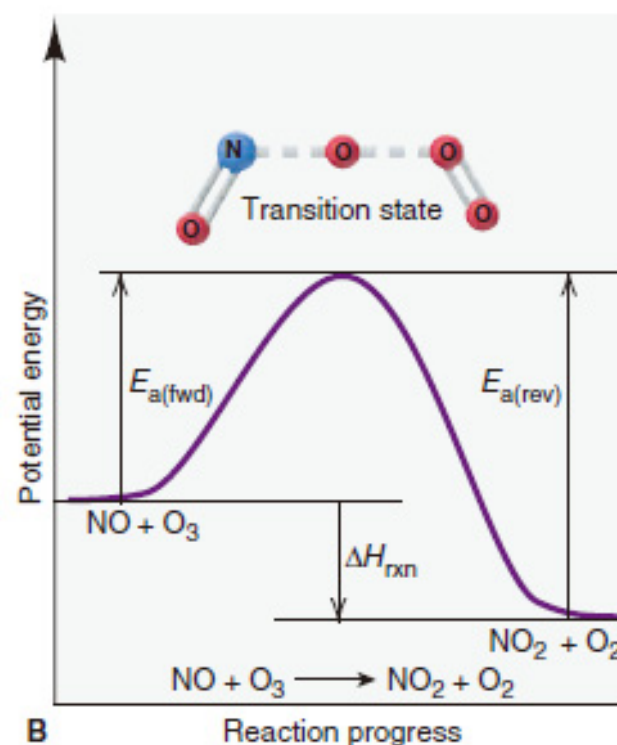
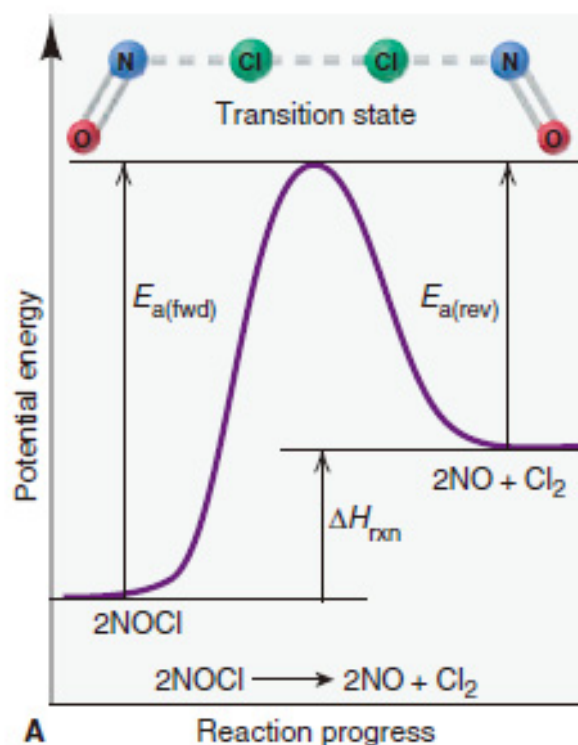
$$\Delta H = E_{a(fwd)} - E_{a(rev)} < 0$$

Exothermic reaction





Transition state theory proposes that every reaction (or every step in an overall reaction) goes through its own transition state.



$$\Delta H = E_{a(\text{fwd})} - E_{a(\text{rev})} > 0$$

Endothermic reaction

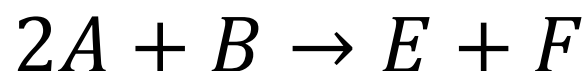
$$\Delta H = E_{a(\text{fwd})} - E_{a(\text{rev})} < 0$$

Exothermic reaction

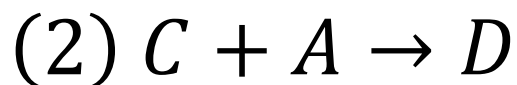
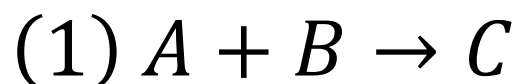


10.6 Reaction mechanisms: the steps from reactant to product

The sequence of single reaction steps that sum to the overall equation.



Might involve these three simpler steps



- Reactions may occur all at once or through several discrete steps.
- Each of these processes is known as an **elementary reaction**.



Elementary reactions and molecularity

Elementary reactions

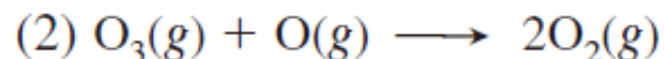
- The individual steps that make up a reaction mechanism.
- An elementary step is characterized by its **molecularity**, the number of reactant particles in the step.



A two-step mechanism has been proposed:



Unimolecular reaction



Bimolecular reaction

- Some termolecular elementary steps occur, but they are extremely rare because the probability of three particles colliding simultaneously with enough energy and an effective orientation is very small.
- Higher molecularities are not known.
- Therefore, in general, we propose unimolecular and/or bimolecular reactions as reasonable steps in a mechanism.



The rate-determining step of a reaction mechanism

Rate-determining step

All the elementary steps in a mechanism have their own rates. However, one step is usually much slower than the others. This step, called the **rate-determining step**, limits how fast the overall reaction proceeds.



If this reaction were an elementary step—that is, if the mechanism consisted of only one step—we could immediately write the overall rate law as

$$\text{Rate} = k[\text{NO}_2][\text{CO}]$$

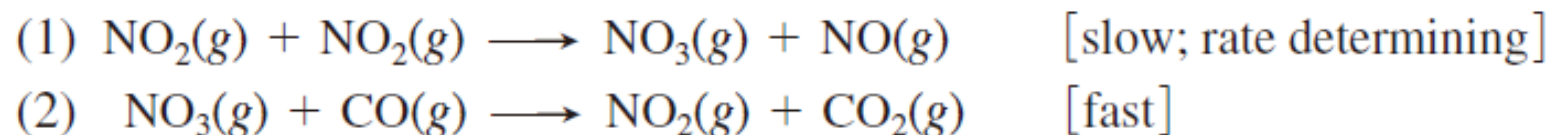
But, if the experimental data show the rate law is

$$\text{Rate} = k[\text{NO}_2]^2$$



Thus, the overall reaction cannot be elementary.

A proposed two-step mechanism is



Reaction intermediate NO_3

- A substance formed and used up during the reaction.
- Intermediates are less stable than the reactants and products, but unlike *much* less stable transition states, they have normal bonds and can sometimes be isolated.



$$(1) \text{ Rate}_1 = k_1[\text{NO}_2][\text{NO}_2] = k_1[\text{NO}_2]^2$$

$$(2) \text{ Rate}_2 = k_2[\text{NO}_3][\text{CO}]$$

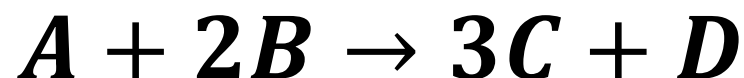
Three key points to notice about this mechanism are:

- If $k = k'$, the rate law for the rate-determining step (step 1) becomes identical to the observed rate law.
- Because the first step is slow, $[\text{NO}_3]$ is low. As soon as any NO_3 forms, it is consumed by the fast second step, so the reaction takes as long as the first step does.
- CO does not appear in the rate law (reaction order = 0) because it takes part in the mechanism after the rate-determining step.



Summary

1. Reaction rate (*Exercise 25.5 a and 25.6a*)



$$v = \frac{d[D]}{dt} = \frac{1}{3} \frac{d[C]}{dt} = -\frac{d[A]}{dt} = -\frac{1}{2} \frac{d[B]}{dt}$$

$$v = \frac{1}{v_j} \times \frac{1}{V} \frac{dn_j}{dt} = \frac{1}{v_j} \frac{d[J]}{dt}$$

$$\text{Unit: mol dm}^{-3} \text{s}^{-1}$$



25.6a

25.6 (a) The rate of formation of C in the reaction $2A + B \rightarrow 2C + 3D$ is $1.0 \text{ mol L}^{-1} \text{ s}^{-1}$. State the reaction rate, and the rates of formation or consumption of A, B, and D.

$$v = \frac{1}{3} \frac{d[D]}{dt} = \frac{1}{2} \frac{d[C]}{dt} = -\frac{1}{2} \frac{d[A]}{dt} = -\frac{d[B]}{dt}$$

$$\frac{d[C]}{dt} = 1.0 \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$$

$$v = 0.5 \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$$

$$\frac{d[A]}{dt} = -1.0 \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1} \quad \text{The rate of consumption of A is } 1.0 \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$$

$$\frac{d[B]}{dt} = -0.5 \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1} \quad \text{The rate of consumption of B is } 0.5 \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$$

$$\frac{d[D]}{dt} = 1.5 \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1} \quad \text{The rate of formation of D is } 1.5 \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$$



2. Rate law (*Self-test 10.1, Exercise 25.7 b, 25.8b, and 25.9b*)

$$v = k[A]^m [B]^n \dots$$

k : rate constant

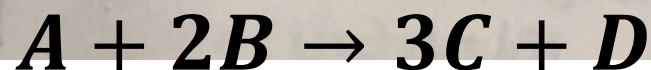
- Specific for a given reaction at a given temperature: Dependent on T
- Does not change as the reaction proceeds.
- Be found by experiment.

m, n : reaction orders



25.7b

25.7 (b) The rate law for the reaction in Exercise 25.5b was found to be $v = k[A][B]^2$. What are the units of k ? Express the rate law in terms of the rates of formation and consumption of (a) A, (b) C.



$$k = \frac{v}{[A][B]^2} = \frac{\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}}{(\text{mol} \cdot \text{L}^{-1})^3} = \text{mol}^{-2} \cdot \text{L}^2 \cdot \text{s}^{-1}$$

$$-\frac{d[A]}{dt} = k[A][B]^2 \quad \Rightarrow \quad \frac{d[A]}{dt} = -k[A][B]^2$$

$$\frac{1}{3} \frac{d[C]}{dt} = k[A][B]^2 \quad \Rightarrow \quad \frac{d[C]}{dt} = 3k[A][B]^2$$



25.9b

25.9 (b) At 400 K, the rate of decomposition of a gaseous compound initially at a pressure of 12.6 kPa, was 9.71 Pa s^{-1} when 10.0 per cent had reacted and 7.67 Pa s^{-1} when 20.0 per cent had reacted. Determine the order of the reaction.

Reactant

Reactant	
$[A]_0 = 12.6 \text{ kPa}$	
$[A]_1/[A]_0 = 0.9$	$v_1 = 9.71 \text{ Pa} \cdot \text{s}^{-1}$
$[A]_2/[A]_0 = 0.8$	$v_2 = 7.67 \text{ Pa} \cdot \text{s}^{-1}$

$$v = k[A]^x$$

$$\frac{v_1}{v_2} = \frac{k[A]_1^x}{k[A]_2^x} = \left\{ \frac{[A]_1}{[A]_2} \right\}^x \quad \Rightarrow \quad x = \ln \frac{\frac{v_1}{[A]_1}}{\frac{v_2}{[A]_2}} = 2$$



3. Integrated Rate law (*Self-test 10.2 and 10.3; Exercise 25.10 b*)

Order	Reaction	Rate law	Integrated rate law	$t_{1/2}$
0	$A \rightarrow P$	$v = k$	$[A] = [A]_0 - kt$	$\frac{[A]_0}{2k}$
1	$A \rightarrow P$	$v = k[A]$	$\ln\left(\frac{[A]}{[A]_0}\right) = -kt$ $[A] = [A]_0 e^{-kt}$	$\frac{\ln 2}{k} = \frac{0.693}{k}$
2	$A \rightarrow P$	$v = k[A]^2$	$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$ $[A] = \frac{[A]_0}{1 + kt[A]_0}$	$\frac{1}{k[A]_0}$
	$A + B \rightarrow P$	$v = k[A][B]$	$\ln\left(\frac{[B]/[B]_0}{[A]/[A]_0}\right) = ([B]_0 - [A]_0)kt$	

$$\int \frac{1}{x} dx = \ln x \quad \int \frac{1}{x^2} dx = -\frac{1}{x}$$



A zero order reaction starts with an initial concentration of reactant of $1.00 \text{ mol} \cdot \text{L}^{-1}$ and has a rate constant of $1.65 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$. What is the concentration of the reactant after (a) 100 s and (b) 100 minutes?

$$v = k$$

$$[A] = [A]_0 - kt$$

$$[A]_0 = 1 \text{ mol} \cdot \text{L}^{-1} \quad k = 1.65 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$$

After 100 s

$$\begin{aligned} [A] &= [A]_0 - kt = 1 \text{ mol} \cdot \text{L}^{-1} - 1.65 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1} \times 100 \text{ s} \\ &= 0.998 \text{ M} \end{aligned}$$

After 100 min = 6000 s

$$\begin{aligned} [A] &= [A]_0 - kt = 1 \text{ mol} \cdot \text{L}^{-1} - 1.65 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1} \times 6000 \text{ s} \\ &= 0.901 \text{ M} \end{aligned}$$



A first order reaction $ABC \rightarrow AB + C$ has a rate constant of 0.0125 s^{-1} . If the initial concentration of ABC is $0.500 \text{ mol} \cdot \text{L}^{-1}$, what is the concentration of ABC after 5 minutes?

$$\ln \left(\frac{[A]}{[A]_0} \right) = -kt \quad \Rightarrow \quad \text{求}k\text{时用}$$

$$[A] = [A]_0 e^{-kt} \quad \Rightarrow \quad \text{求}[A]\text{时用}$$

$$\begin{aligned} [A] &= [A]_0 e^{-kt} = 0.500 \text{ mol} \cdot \text{L}^{-1} \times e^{(-0.0125 \text{ s}^{-1} \times 300 \text{ s})} \\ &= 0.0118 \text{ mol} \cdot \text{L}^{-1} \end{aligned}$$



A reaction $X_2Y \rightarrow 2X + Y$ has a rate law:

$$\text{Rate} = 0.104 \text{ (mol}^{-1}\cdot\text{L}\cdot\text{s}^{-1})\cdot[X_2Y]^2.$$

If the initial concentration of products is zero, how long does it take for the concentration of X to reach $0.100 \text{ mol}\cdot\text{L}^{-1}$ starting with an initial concentration of X_2Y of $0.200 \text{ mol}\cdot\text{L}^{-1}$?

	X_2Y	$\rightarrow$	$2X + Y$
Initial	0.200 M		0 M
Final	$[X_2Y] = 0.200 - \frac{1}{2} \times 0.1$ $= 0.150 \text{ M}$		0.1 M

$$\text{Rate} = 0.104 \text{ (mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1})\cdot[X_2Y]^2 \quad \text{Second-order reaction}$$

$$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$$

$$\frac{1}{0.150 \text{ mol}\cdot\text{L}^{-1}} - \frac{1}{0.200 \text{ mol}\cdot\text{L}^{-1}} = 0.104 \text{ mol}^{-1}\cdot\text{L}\cdot\text{s}^{-1} \times t$$

$$t = 16.0 \text{ s}$$



4. Arrhenius Equation (*Self-test 10.4; Exercise 25.16 b*)

a. Definition

$$k = Ae^{-E_a/RT}$$

E_a : activation energy

A: frequency factor

$$A = pZ$$

Z: collision frequency

p: an orientation probability factor; The more complex the molecular structure, the smaller the p value is.

b. Calculation

$$\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



Self test 10.4

The rate constant for the formation of hydrogen iodide from its elements is $2.7 \times 10^{-4} \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$ at 600 K and $3.5 \times 10^{-3} \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$ at 650 K. Find the activation energy, E_a .



$$\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\ln \frac{3.5 \times 10^{-3}}{2.7 \times 10^{-4}} = -\frac{E_a}{8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}} \left(\frac{1}{650} - \frac{1}{600} \right)$$

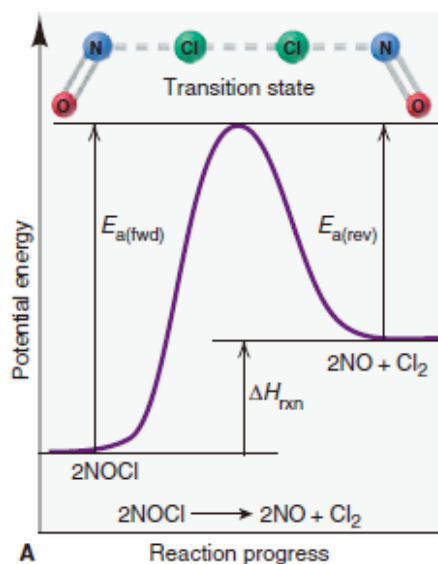
$$E_a = 1.66 \times 10^5 \text{ J}$$



c. Reaction energy diagram

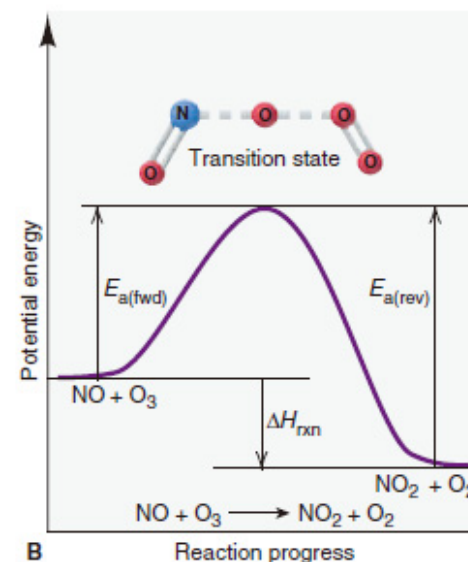
 $E_{a(fwd)}$ and $E_{a(rev)}$ and transition state

- $E_{a(fwd)}$, is the energy difference between the activated state and the reactants;
- $E_{a(rev)}$, is the energy difference between the activated state and the products.



$$\Delta H = E_{a(fwd)} - E_{a(rev)} > 0$$

Endothermic reaction



$$\Delta H = E_{a(fwd)} - E_{a(rev)} < 0$$

Exothermic reaction



5. Reaction mechanisms

a. Elementary reactions and molecularity

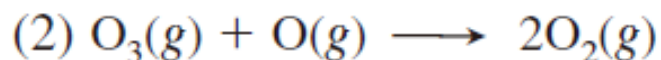
- In general, only unimolecular and/or bimolecular reactions are reasonable steps in a mechanism.
- Reaction intermediate (having normal bonds and can sometimes be isolated.)



A two-step mechanism has been proposed:



Unimolecular reaction



Bimolecular reaction



b. Rate-determining step

$$k_{overall} = k_{slow}$$

A valid mechanism must meet three criteria:

1. The elementary steps must add up to the overall balanced equation.
2. The elementary steps must be reasonable. They should generally involve one reactant particle (unimolecular) or two (bimolecular).
3. The mechanism must correlate with the rate law, not the other way around.